Question	Answer	Marks
12(a)	unstable nucleus	**B1**
	emission of particles/photons	**B1**
	emission is spontaneous **or** (particles/radiation) are ionising	**B1**
12(b)(i)	tangent drawn and gradient calculation attempted	**B1**
	activity = 1.3×10^{6} Bq (*1 mark for answer within* $\pm 0.2 \times 10^{6}$ Bq*, 2 marks for answer within* $\pm 0.1 \times 10^{6}$ Bq)	**A2**
12(b)(ii)	$A = \lambda N$	**C1**
	$\lambda = (1.3 \times 10^{6}) / (3.05 \times 10^{10}) = 4.3 \times 10^{-5}\,\text{s}^{-1}$ ($\approx 4 \times 10^{-5}\,\text{s}^{-1}$)	**A1**
12(c)	$A = A_0 e^{-\lambda t}$ $1.0 \times 10^{3} = 4.6 \times 10^{3} \exp(-5.5 \times 10^{-7} \times t)$	**C1**
	$\ln(4.6) = 5.5 \times 10^{-7} \times t$	**C1**
	$t = 2.78 \times 10^{6}$ s = 32 days	**A1**